Predictive Maintenance for Industrial Cooling Pumps

Hybrid ML approach using Isolation Forest & Linear Regression

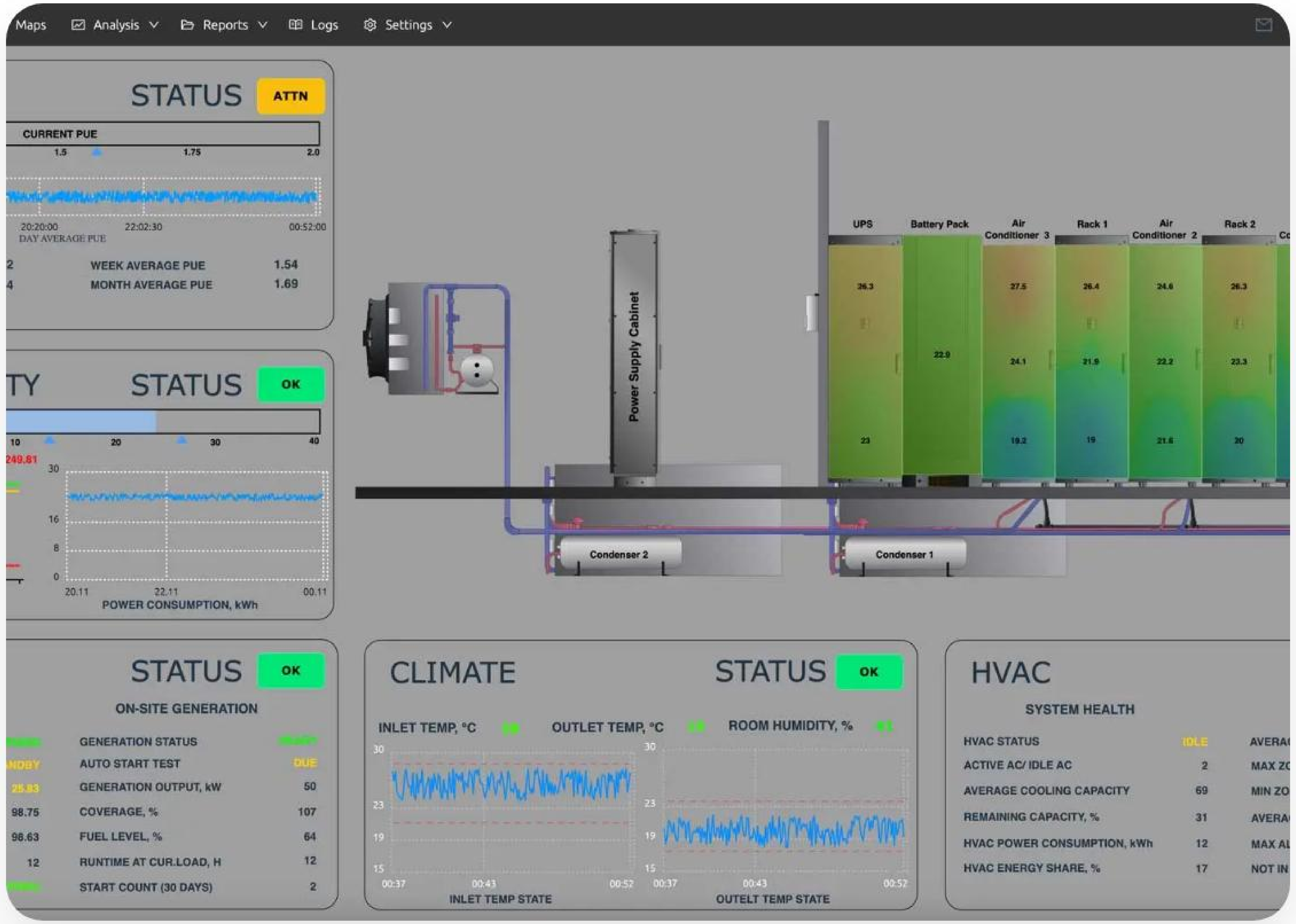
Course: CU_Project | Edge AI Implementation

SYSTEM OVERVIEW

The Industrial Challenge

Standard SCADA systems rely on static thresholds, often missing the subtle "drift" of mechanical degradation. Our solution employs a **"Belt and Suspenders"** methodology.

- **Deterministic Layer:** Statistical Z-Score overrides for sudden spikes.
- **Probabilistic Layer:** Unsupervised Isolation Forest for multi-vector anomaly detection.
- **Prognostic Layer:** Linear Regression for real-time RUL forecasting.



USE CASE ANALYSIS

UC-01: ML Training

Initializes "Machine Memory" from SQLite history; fits Isolation Forest model on healthy operational data baseline.

UC-02: Evaluation

Real-time ingestion of telemetry to assess machine state against unsupervised ML boundaries.

UC-03: RUL Predict

Activates supervised Linear Regression upon anomaly detection to project Remaining Useful Life based on degradation velocity.

Primary Actors: Sensor Module (System), Edge AI Node (System), Maintenance Engineer (Human).

HYBRID ANOMALY DETECTION

A. Isolation Forest

Identifies multi-vector outliers by isolating data points through decision trees. Anomalies require fewer splits.

Hyperparameter: *contamination=0.02* (2% historical noise assumption).

B. Z-Score Override

Parallel statistical math to catch rapid linear degradations and prevent "drift blindness."

$$Z = \frac{X - \mu}{\max(\sigma, 0.5)}$$

Override Threshold: **$|Z| > 3.5$**

Centrifugal Pump

RUL FORECASTING

Linear Regression Engine

Once a fault is latched (3-strike persistence), the system buffers the last 10 seconds of data.

$$\text{RUL} = \frac{\text{SafetyLimit} - \text{CurrentValue}}{\text{ProjectedVelocity}}$$

The **slope** of the regression hyperplane determines the degradation velocity, providing a deterministic countdown to the NEMA/ISO safety limit.

FINITE STATE MACHINE (FSM)

State	Name	Logic & Trigger Condition
State 0	Initialization	Boots MQTT, trains baselines, applies Poison Filter to SQLite history.
State 1	Rest/Monitoring	Nominal operation. Consecutive anomaly counter is zero.
State 2	Drift Identification	Unverified anomaly (Strike 1 or 2). System holds for confirmation.
State 3	Stage 1 Warning	Latched alarm (Strike 3+). RUL Engine active. Digital Twin alerts.
State 4	Stage 2 Critical	Hard Limit (e.g. >80°C) breached. Emergency Shutdown (ESD) issued.
State 5	Repair Validation	Hysteresis logic: Requires 3 consecutive healthy readings to return to State 1.

UML SEQUENCE INTERACTION

1. Perception

Sensor Simulator publishes Raw Telemetry (JSON) to Mosquitto Broker at 1 Hz.

2. Intelligence

Edge AI Node parses payload, applies Smoothing, runs Hybrid Inference, and logs to SQLite.

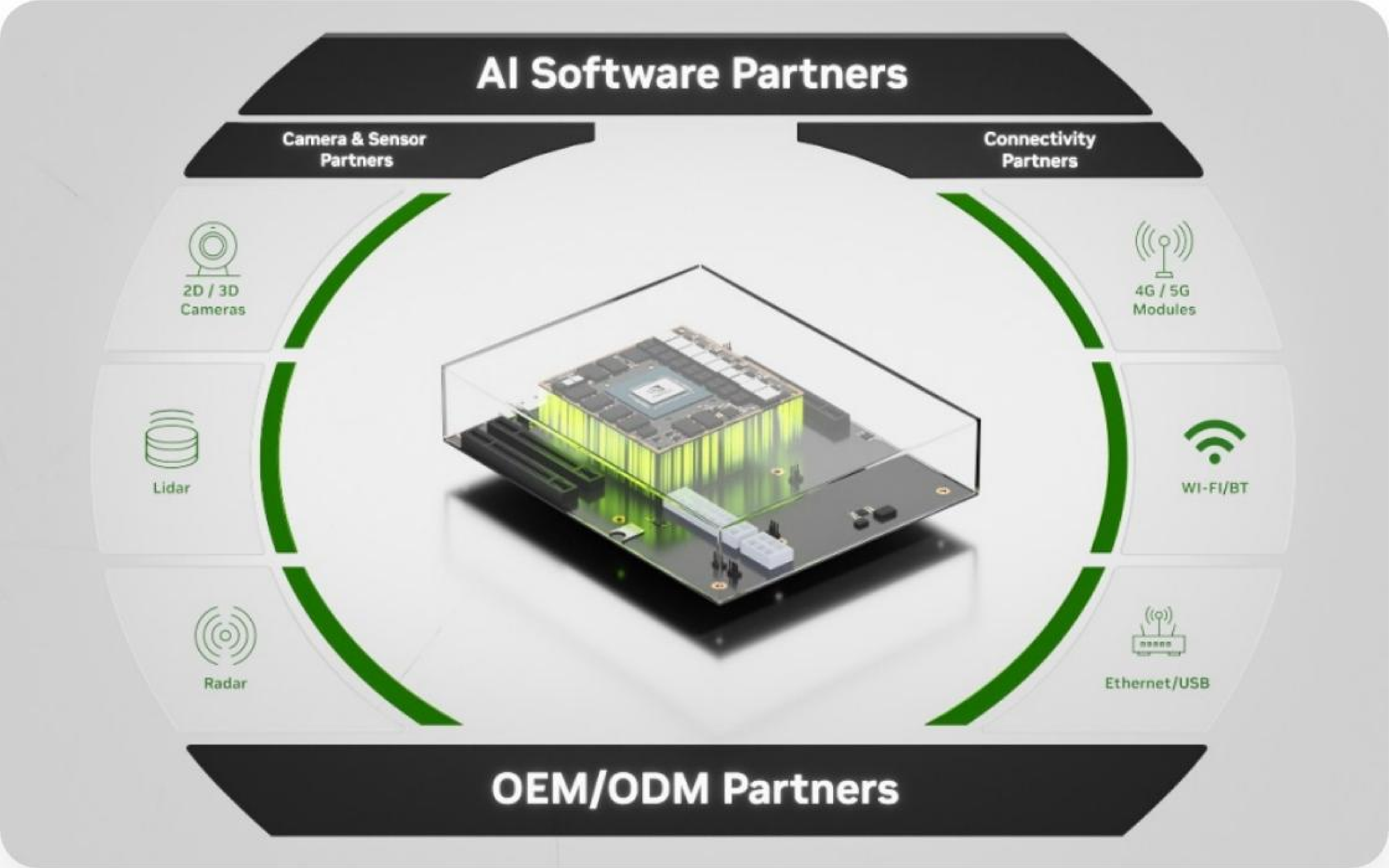
3. Presentation

Streamlit Dashboard polls SQLite asynchronously, renders Plotly charts and RUL countdown.

→

Latency Optimized: End-to-end processing < 15ms.

EDGE INFRASTRUCTURE



Target Environment

Optimized for ARM Cortex-A72 architectures (Raspberry Pi 4 / Jetson Nano).

- **Processor:** Quad-core 1.5 GHz+ (supports twin inference engines).
- **Memory:** 2 GB LPDDR4 minimum (efficient Python/Mosquitto footprint).
- **Storage:** Localized SQLite for high-availability "Machine Memory."

JSON MESSAGING SCHEMA

Diagnostic Payload

Upon fault identification, the system dynamically generates Root-Cause Analysis (RCA) metadata.

- Status & Message tags.
- Faulty Sensor ID.
- Prognostic Confidence markers.


```
{ "status": "CRITICAL", "message": "LIMIT EXCEEDED: temperature", "prognostics": { "confidence_pct": 94.5, "estimated_rul_sec": 42 }, "sensor_data": { "vibration_rms": 3.102, "faulty_sensor": "temperature" } }
```

VALIDATION & PERFORMANCE

Unit Testing

Verified +1.0°C/sec slope detection and Gaussian noise filtering (< 2% variance).

Integration

Validated concurrent SQL read/write and 1 Hz MQTT throughput with zero database lockout.

Performance

Avg Latency: **8-15ms** (Exceeding NFR-01 target of 50ms).

DEPLOYMENT MANUAL

1. **Environment:** Run `pip install -r requirements.txt` to setup ML and MQTT frameworks.
2. **Connectivity:** Start Mosquitto Broker on Port 1883.
3. **Simulation:** Launch `python cooling_pump_sim.py` (1 Hz telemetry).
4. **Intelligence:** Launch `python edge_node.py` (Isolation Forest / Z-Score).
5. **Digital Twin:** Launch `streamlit run dashboard.py` for real-time visualization.

Master Command: `python run_all.py`

REFERENCES & CONCLUSION

[1] ISO 20816-1:2017 Mechanical vibration standards.

[2] NEMA MG 1-2021: Motors and Generators guidelines.

[3] Liu et al. (2008) "Isolation Forest" IEEE ICDM.

[4] NIST SP 800-82 Revision 2: ICS Security.

[5] Si et al. (2011) "Remaining useful life estimation" EJOR.

Questions?

Thank you for your attention.

Predictive Maintenance Project | CU_Project

github.com/smarthsalaria/predictive_maintenance

IMAGE SOURCES

https://www.cuchd.in/about/assets/images/cu-logo.png

Source: www.cuchd.in

Thumbnail for centosoftware.com

https://centosoftware.com/wp-content/uploads/2025/11/1920-1000-CENTO-dashboard-DCIM.webp

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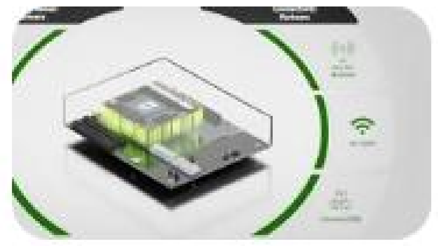

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